**Shiv Nadar University Chennai**

M.Tech. Artificial Intelligence and Data Science

**Deep Learning Laboratory (26AI5705)**

# Deep Learning Laboratory Report

Lab Assignments 1 and 2

**Team 17**

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# 1 Assignment 1

## 1.1 Objective

The assignment covers vector comparison, activation functions and derivatives, one feedforward/backpropagation step, and feature normalization. MATLAB source files are provided in the `lab1` folder.

## 1.2 Vector Distance and Similarity

For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$,

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{\sum_{i=1}^{n} (u_i - v_i)^2}, \tag{1.1}$$

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}^T \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}. \tag{1.2}$$

Using

$$\mathbf{x} = [3, -4, 6, -1], \quad \mathbf{y} = [-2, 5, 8, -3], \quad \mathbf{z} = [5, 3, -1, -4],$$

the results are:

Table 1.1: Vector comparison results.

| Pair | Euclidean distance | Cosine similarity |
|---|---|---|
| $\mathbf{x}, \mathbf{y}$ | 10.6771 | 0.3144 |
| $\mathbf{x}, \mathbf{z}$ | 10.5357 | 0.0178 |

The first pair has the larger angular similarity. The second pair is almost orthogonal.

## 1.3 Activation Functions

Table 1.2: Activation functions used in the assignment.

| Function | $f(x)$ | Output range |
|---|---|---|
| Sigmoid | $\dfrac{1}{1+e^{-\beta x}}$ | $(0, 1)$ |
| Tanh | $\tanh(\beta x)$ | $(-1, 1)$ |
| ReLU | $\max(0, x)$ | $[0, \infty)$ |
| Softplus | $\ln(1 + e^x)$ | $(0, \infty)$ |
| Leaky ReLU | $\begin{cases} \delta x, & x < 0 \\ x, & x \geq 0 \end{cases}$ | $(-\infty, \infty)$ for $\delta > 0$ |
| Swish | $x \,\mathrm{sigmoid}(\beta x)$ | approximately $[-0.278, \infty)$ for $\beta = 1$ |

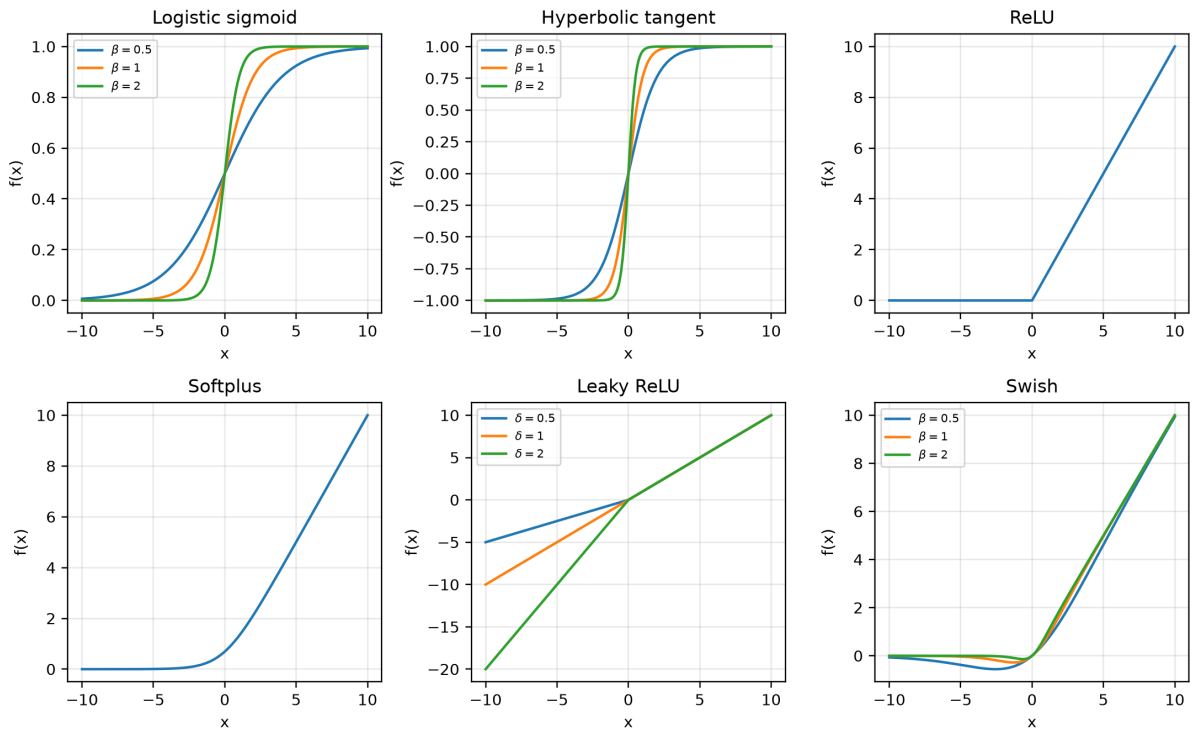



Figure 1.1: Activation functions for the required $\beta$ and $\delta$ values.

Increasing $\beta$ makes sigmoid and tanh steeper. ReLU is piecewise linear, Softplus is its smooth approximation, and Swish is smooth and non-monotonic.

## 1.4 Activation Derivatives

Table 1.3: Analytical derivatives. Values at $x = 0$ for piecewise functions may be assigned by convention.

| Function | $f'(x)$ |
|---|---|
| Sigmoid | $\beta\sigma(\beta x)\big(1 - \sigma(\beta x)\big)$ |
| Tanh | $\beta\big(1 - \tanh^2(\beta x)\big)$ |
| ReLU | 0 for $x < 0$; 1 for $x > 0$ |
| Softplus | $\sigma(x)$ |
| Leaky ReLU | $\delta$ for $x < 0$; 1 for $x > 0$ |
| Swish | $\sigma(\beta x) + \beta x\sigma(\beta x)\big(1 - \sigma(\beta x)\big)$ |

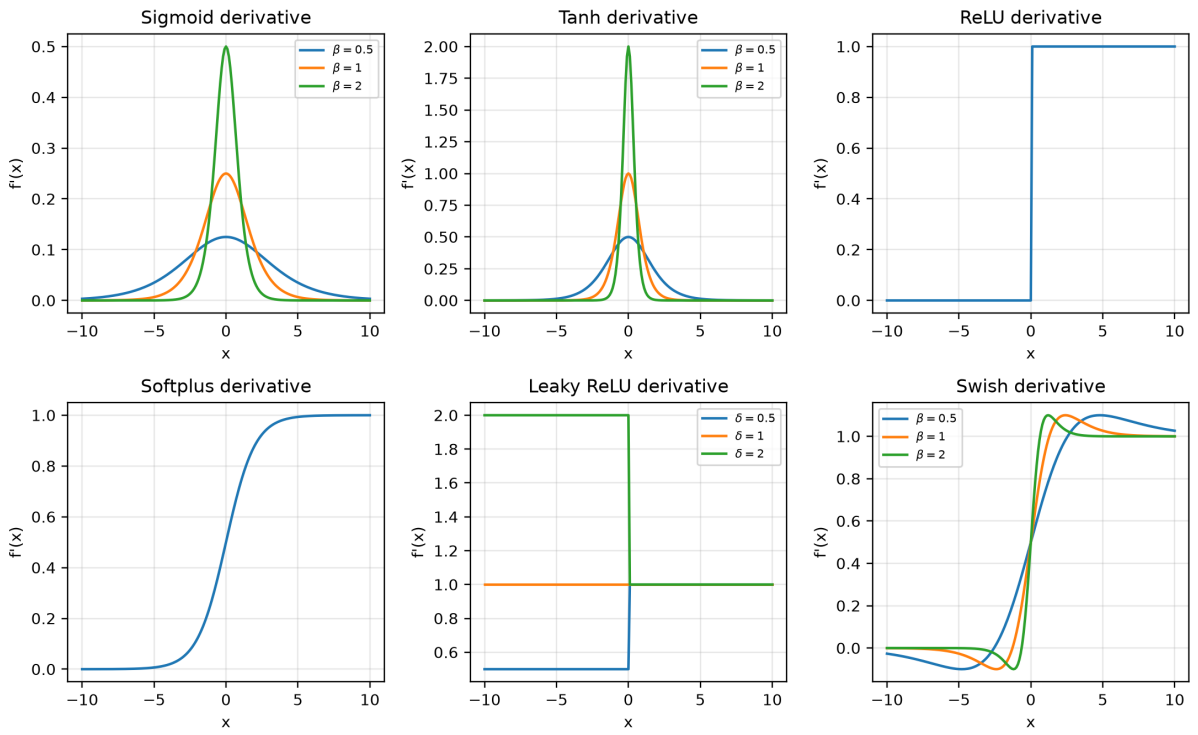



Figure 1.2: Derivatives of the six activation functions.

## 1.5 Feedforward and Backpropagation

The network contains two inputs, four tanh hidden nodes, and three sigmoid output nodes. The Class 2 target is

$$\mathbf{x} = [2.5, -4.3], \qquad \mathbf{t} = [0, 1, 0], \qquad \eta = 0.9.$$

### 1.5.1 Forward pass

$$\mathbf{a}^h = \mathbf{x}W_h + \mathbf{b}_h, \qquad \mathbf{h} = \tanh(\mathbf{a}^h), \tag{1.3}$$

$$\mathbf{a}^o = \mathbf{h}W_o + \mathbf{b}_o, \qquad \mathbf{y} = \text{sigmoid}(\mathbf{a}^o), \tag{1.4}$$

$$E = \frac{1}{2}\sum_{k=1}^{3}(t_k - y_k)^2. \tag{1.5}$$

$$\begin{aligned}
\mathbf{a}^h &= [-5.0900,\ 3.8600,\ 0.4100,\ -7.6700],\\
\mathbf{h} &= [-0.999924,\ 0.999113,\ 0.388473,\ -1.000000],\\
\mathbf{a}^o &= [2.010487,\ 1.255786,\ 1.775349],\\
\mathbf{y} &= [0.881894,\ 0.778300,\ 0.855122],\\
E &= 0.779060.
\end{aligned}$$

### 1.5.2 Local gradients

$$\boldsymbol{\delta}^o = (\mathbf{t} - \mathbf{y}) \odot \mathbf{y} \odot (1 - \mathbf{y}), \tag{1.6}$$

$$\boldsymbol{\delta}^h = (\boldsymbol{\delta}^o W_o^T) \odot (1 - \mathbf{h}^2). \tag{1.7}$$

$$\boldsymbol{\delta}^o = [-0.091856,\ 0.038254,\ -0.105940],$$
$$\boldsymbol{\delta}^h = [2.8591\times10^{-5},\ -3.0012\times10^{-4},\ -0.324714,\ -2.5054\times10^{-7}].$$

### 1.5.3 Parameter changes

$$\Delta W_o = \eta\, \mathbf{h}^T \boldsymbol{\delta}^o, \qquad \Delta \mathbf{b}_o = \eta \boldsymbol{\delta}^o, \tag{1.8}$$
$$\Delta W_h = \eta\, \mathbf{x}^T \boldsymbol{\delta}^h, \qquad \Delta \mathbf{b}_h = \eta \boldsymbol{\delta}^h. \tag{1.9}$$

$$\Delta W_o = \begin{bmatrix} 0.082664 & -0.034426 & 0.095339 \\ -0.082597 & 0.034398 & -0.095261 \\ -0.032115 & 0.013375 & -0.037039 \\ 0.082670 & -0.034429 & 0.095346 \end{bmatrix}, \quad \Delta \mathbf{b}_o = [-0.082670,\ 0.034429,\ -0.095346].$$

$$\Delta W_h = \begin{bmatrix} 0.000064 & -0.000675 & -0.730605 & -0.000001 \\ -0.000111 & 0.001161 & 1.256641 & 0.000001 \end{bmatrix},$$
$$\Delta \mathbf{b}_h = [0.000026,\ -0.000270,\ -0.292242,\ -0.000000].$$

The updated parameters are obtained using $W^{\text{new}} = W + \Delta W$ and $\mathbf{b}^{\text{new}} = \mathbf{b} + \Delta \mathbf{b}$.

## 1.6 Feature Normalization

Min-max and mean-variance normalization are

$$x_i' = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}, \tag{1.10}$$
$$z_i = \frac{x_i - \bar{x}}{s}. \tag{1.11}$$

MATLAB's sample standard deviation is used: $\bar{x}_1 = 1.3$, $s_1 = 4.5717$, $\bar{x}_2 = 2.1$, and $s_2 = 9.9605$.

Table 1.4: Normalized feature values.

| $x_1$ | $x_2$ | Min-max $x_1$ | Min-max $x_2$ | Standard $x_1$ | Standard $x_2$ |
|---|---|---|---|---|---|
| -2 | 12 | 0.2857 | 0.8929 | -0.7218 | 0.9939 |
| 5 | 15 | 0.7857 | 1.0000 | 0.8093 | 1.2951 |
| -3 | -13 | 0.2143 | 0.0000 | -0.9406 | -1.5160 |
| 7 | -6 | 0.9286 | 0.2500 | 1.2468 | -0.8132 |
| 4 | 4 | 0.7143 | 0.6071 | 0.5906 | 0.1908 |
| -1 | -7 | 0.3571 | 0.2143 | -0.5031 | -0.9136 |
| 8 | 13 | 1.0000 | 0.9286 | 1.4656 | 1.0943 |
| -6 | -2 | 0.0000 | 0.3929 | -1.5968 | -0.4116 |
| 0 | 10 | 0.4286 | 0.8214 | -0.2844 | 0.7931 |
| 1 | -5 | 0.5000 | 0.2857 | -0.0656 | -0.7128 |

## 1.7 Conclusion

The calculations verify the vector metrics, activation derivatives, one generalized-delta update, and both normalization methods. The figures show how the activation parameters control slope and gradient magnitude.

# 2 Assignment 2

## 2.1 Objective

The task is to classify two nonlinearly separable classes using two hidden-layer networks. The 6-4 and 12-8 configurations are compared using validation accuracy, and the selected model is evaluated on the test set.

## 2.2 Dataset and Preprocessing

Dataset 1 contains two features and a binary label.

Table 2.1: Dataset splits.

| Split | Examples | Use |
|---|---|---|
| Training | 1123 | Weight updates |
| Validation | 317 | Configuration selection |
| Test | 160 | Final evaluation |

Each feature is standardized using training statistics only:

$$x'_j = \frac{x_j - \mu_j^{\text{train}}}{s_j^{\text{train}}}. \tag{2.1}$$

This prevents validation and test information from entering training.

## 2.3 Network Model

Table 2.2: Network configurations.

| Configuration | Inputs | Hidden layer 1 | Hidden layer 2 | Output |
|---|---|---|---|---|
| 6-4 | 2 | 6 | 4 | 1 |
| 12-8 | 2 | 12 | 8 | 1 |

For one input pattern $\mathbf{x}$,

$$\mathbf{h}_1 = \tanh(W_1\mathbf{x} + \mathbf{b}_1), \tag{2.2}$$
$$\mathbf{h}_2 = \tanh(W_2\mathbf{h}_1 + \mathbf{b}_2), \tag{2.3}$$
$$\hat{y} = \text{sigmoid}(W_3\mathbf{h}_2 + b_3). \tag{2.4}$$

The activations use $\beta = 1$. Binary cross-entropy is

$$E = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \ln(\hat{y}_i) + (1 - y_i)\ln(1 - \hat{y}_i)\right]. \tag{2.5}$$

## 2.4 Training Method

Pattern-mode learning updates the parameters after every example. SGD uses the required learning rate and momentum:

$$\Delta\theta_t = -\eta\nabla_\theta E_t + \alpha\Delta\theta_{t-1}, \qquad \eta = 0.8, \quad \alpha = 0.9. \tag{2.6}$$

The patterns are shuffled each epoch, and the seed is reset before each configuration. Gradient norm clipping at 0.1 limits unstable single-pattern updates. Training stops after three consecutive epochs satisfy

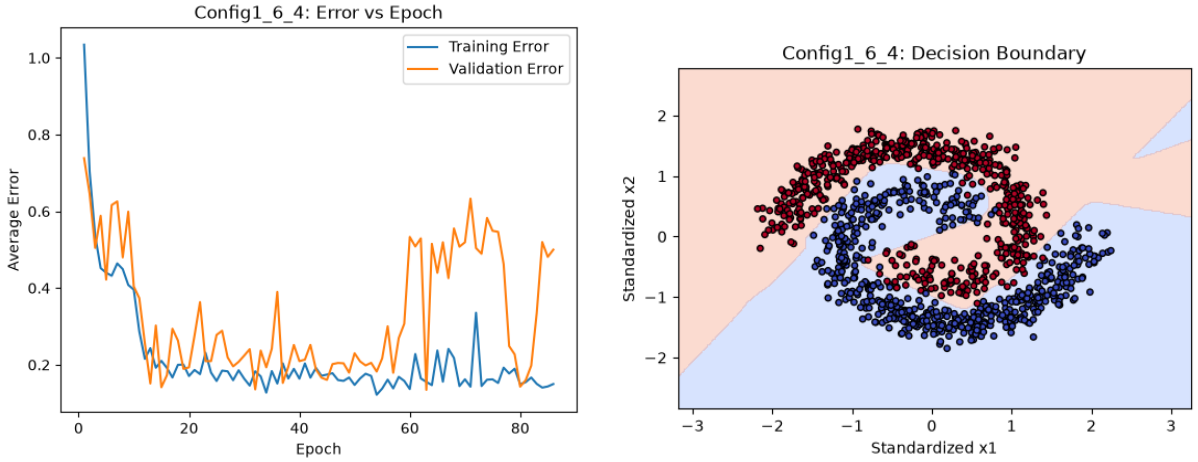
$$\left|E_{\text{train}}^{(e)} - E_{\text{train}}^{(e-1)}\right| < 0.01. \tag{2.7}$$

The three-epoch check avoids stopping because of one noisy pattern-mode epoch.

## 2.5 Results

### 2.5.1 Configuration 1: 6-4

Training stopped after 86 epochs. The final training accuracy was 97.60% and validation accuracy was 96.21%.

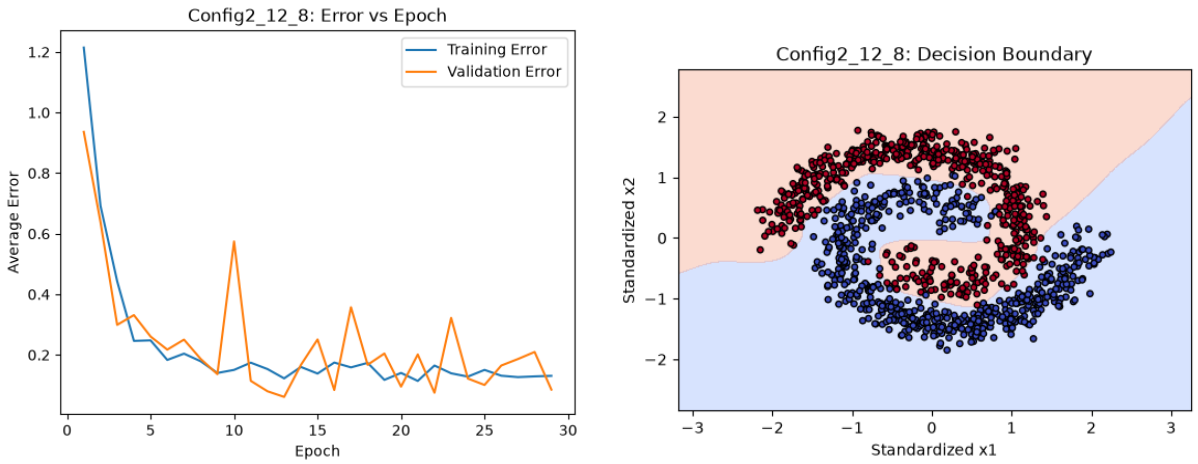



(a) Training and validation error (b) Decision boundary

Figure 2.1: Results for the 6-4 configuration.

### 2.5.2 Configuration 2: 12-8

Training stopped after 29 epochs. The final training accuracy was 98.13% and validation accuracy was 97.79%.




(a) Training and validation error (b) Decision boundary

Figure 2.2: Results for the 12-8 configuration.

### 2.5.3 Validation Selection

Table 2.3: Accuracy comparison.

| Configuration | Stop epoch | Training accuracy | Validation accuracy |
|---|---|---|---|
| 6-4 | 86 | 97.60% | 96.21% |
| 12-8 | 29 | 98.13% | 97.79% |

The 12-8 configuration has the higher validation accuracy and is selected. Its test accuracy is

$$\boxed{98.75\%}. \tag{2.8}$$

## 2.6 Discussion

The error curves fluctuate because pattern-mode updates use one shuffled example at a time with a large learning rate and momentum. The training-validation accuracy gaps are only 1.39 and 0.34 percentage points, so there is no strong evidence of overfitting. Both decision plots capture the nonlinear spiral-like class boundary.

## 2.7 Conclusion

Both networks learned the nonlinear classes. Validation accuracy selected the 12-8 model, which achieved 98.75% accuracy on the unseen test data.